Revised Evaluation and Results

Spearman correlation and summed proper log score

October 26, 2026

4.3 Evaluation

The analysis unit is a context–trigger–query combination in the focus study, a lexical scale within each Hu et al. (2023) benchmark dataset, and a scale–condition combination in Ronai and Xiang (2024). We follow the inclusion subset reproduced from Hu et al.'s original analysis code, giving 39 van Tiel et al. (2016) scales, 67 Gotzner et al. (2018) scales, 50 Pankratz and van Tiel (2021) scales, and 57 Ronai and Xiang (2022) scales. For van Tiel et al., we average each predictor's probabilities across the three sentence templates before comparison with the scale's human inference rate. Together with 480 focus pairs and 300 Ronai and Xiang (2024) scale–condition observations, this yields 993 analysis units.

We use two complementary evaluation measures: Spearman correlation for agreement about the relative ordering of exclusion judgments, and summed proper log score for probability prediction. For human exclusion rate y_i and predicted exclusion probability p_i, the reported log score over analysis units D is

$$L_D = \sum_{i \in D} \left[y_i \log \widetilde{p}_i + (1 - y_i) \log(1 - \widetilde{p}_i) \right], \tag{1}$$

where $\widetilde{p}_i$ clips p_i to $[10^{-10}, 1 - 10^{-10}]$ for scoring. Logarithms are natural, and higher scores indicate better probability predictions. Scores sum item contributions without dividing by the number of items or weighting by participant counts. We compare models on matching item sets within a dataset, condition, or subset, because totals depend on the number of items. The training-fold boundary-selection objective remains the balanced mean Set log score described in Section 3.5; reported sums do not change which boundaries are selected.

For the scalar data, Spearman correlations compare scales within each Hu dataset, after template aggregation, and within each Ronai and Xiang condition. For focus exclusion, we correlate predicted probabilities with human exclusion rates across the 30 ordered trigger–query pairs within each context, then average the defined correlations equally across contexts. Ties receive average ranks. Correlations involving a constant vector are undefined and are omitted from means, with valid-context counts reported. Separately, word-ranking Spearman compares neutral language-model continuation scores for the six tested focus words with their independently normed human ordering in each context. Word-ranking agreement and exclusion-ranking agreement therefore concern different targets.

To examine whether linking gains persist when the tested word rankings agree, we summarize the existing predictions in contexts with word-ranking Spearman equal to 1 or greater than .9. These criteria retain two contexts (60 pairs) and seven contexts (210 pairs), respectively. The subsets overlap, and neither model predictions nor selected boundaries are refitted. For the human experimental summary, we average item exclusion rates within each normed condition and context, then average equally across contexts. Model comparisons and condition summaries are descriptive; no tests of differences between models are reported.

5 Results

We evaluate how contextual expectations contribute to alternative exclusion using Spearman correlation and summed proper log score. The two measures distinguish agreement about exclusion rankings from the probability assigned to exclusion and non-exclusion responses. We first compare probability predictions across datasets, then examine the human focus judgments and the contribution of the explicit linking rules. Within the focus analysis, we ask whether contextual expectations already contain ranking information and whether linking gains persist where model and human word rankings agree. We then consider the scalar benchmarks and the QUD and *only* manipulations.

5.1 Linking approaches improve on raw continuation probabilities

Across every evaluated dataset and condition, at least one explicit linking rule has a higher summed log score than No linking (Table 3). X but not Y also improves on No linking in all four Hu et al. scalar benchmarks. This pattern supports a distinction between predicting which expressions are likely continuations and predicting which alternatives listeners exclude. A candidate can receive only a small share of possible continuations while being excluded by most listeners.

The approaches introduce an exclusion relation at different points in the computation. X but not Y supplies the trigger and an explicit contrast in the language-model prompt. Set, Ordering, Conjunction, and Disjunction instead apply an explicit linking rule to representations constructed from neutral contextual expectations. Following Section 3, we distinguish contrastive conditioning from the four explicit linking rules.

The improvement of at least one linking rule does not mean that every rule improves every dataset. The identity of the strongest approach varies, and some variants perform worse than No linking. We examine the focus results in more detail to identify what the mappings contribute and where their predictions remain limited.

Dataset or condition	N	No linking	X but not Y	Best explicit rule	Log score
van Tiel et al. (2016)	39	−113.229	−62.936	Set Top-p	−48.023
Gotzner et al. (2018)	67	−309.738	−134.909	Disjunction Top-K	−190.235
Pankratz and van Tiel (2021)	50	−137.042	−65.196	Ordering	−61.911
Ronai and Xiang (2022)	57	−174.336	−75.688	Disjunction Top-K	−63.059
R&X (2024): Baseline	60	−183.392	−73.146	Ordering	−68.489
R&X (2024): Weak QUD	60	−107.097	−79.663	Ordering	−102.122
R&X (2024): Strong QUD	60	−81.824	−56.702	Conjunction Top-K	−77.445
R&X (2024): *Only*	60	−333.149	−153.979	Disjunction Top-K	−88.648
R&X (2024): Strong QUD + *only*	60	−107.959	−49.439	Conjunction Top-K	−37.425
Focus alternatives	480	−1544.388	−1847.238	Disjunction Top-K	−729.351

Table 3: Summed item log scores; higher values indicate better probability predictions. Models are compared on the same items within each dataset or condition; totals from different-sized datasets are not directly comparable. The best explicit rule is the highest-scoring observed variant among Set, Ordering, Conjunction, and Disjunction, including Top-K and Top-p variants where applicable. These observed maxima do not evaluate a separate procedure for selecting a rule for a new dataset. Values are descriptive point estimates.

5.2 Focus Alternative Results

Human exclusion judgments. We grouped human judgments according to which alternative structures predict exclusion when applied categorically to the independently normed alternatives described in Section 2.1. Mean exclusion was 0.787 where all four structures predict exclusion and 0.213 where none does (Table 4). The diagnostic cases distinguish Set from Ordering. Exclusion averaged 0.624 where Set and Disjunction predict exclusion but Ordering and Conjunction do not. Conversely, exclusion averaged 0.414 where Ordering and Disjunction predict exclusion but Set and Conjunction do not. We calculated exclusion rates for each trigger–query pair, averaged these equally within each condition and context, and then averaged equally across the 16 contexts. Thus, listeners excluded alternatives in both cases that distinguish Set and Ordering, with a higher mean in the case supported by Set. These graded judgments provide the behavioral target for the probabilistic models evaluated below.

Probability predictions. All four explicit linking rules in the Top-K comparison have higher summed log scores than No linking and X but not Y (Figure 2). Ordering has no set-size parameter and is the same in the Top-K and Top-p comparisons. Disjunction Top-K has the highest log score among the evaluated focus models. Allowing either membership or relative ordering to support exclusion therefore gives the strongest observed probability predictions under this implementation. This comparison does not establish that humans use a disjunctive rule; it identifies a useful combination of the two relations for the evaluated predictions.

The Top-p variants show that fit also depends on how sets are constructed. Disjunction Top-p performs similarly to its Top-K counterpart, whereas Set Top-p and Conjunction Top-p perform worse than No

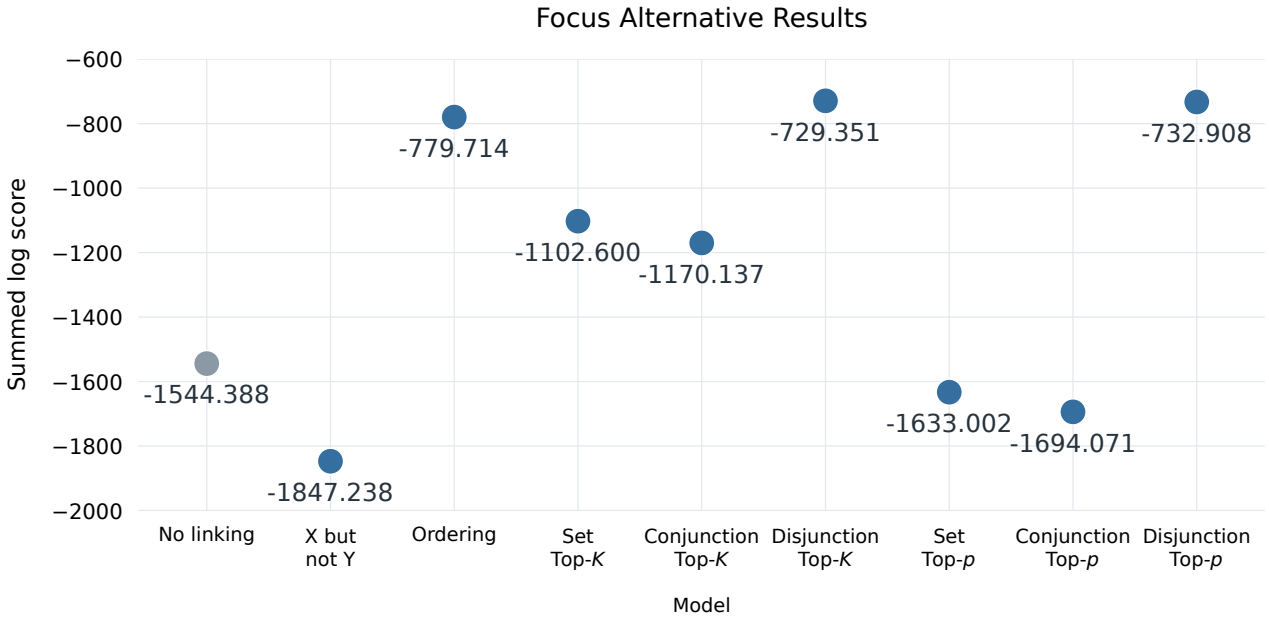


Figure 2: Summed proper log scores for the focus-alternative experiment. Each dot sums the item contributions over the same 480 ordered trigger–query pairs across 16 contexts. Higher (less negative) values indicate better probability predictions. The figure includes No linking, X but not Y, Ordering, and the Top-K and Top-p variants of Set, Conjunction, and Disjunction. Ordering has no set-size parameter. Scores are not divided by item or context count and are not participant-count weighted. Values are descriptive point estimates; uncertainty intervals and significance tests are not shown.

Structures predicting exclusion	Exclusion	Pairs
All four	0.787	192
Set, Disjunction	0.624	48
Ordering, Disjunction	0.414	48
None	0.213	192

Table 4: Human exclusion grouped by the categorical predictions of the alternative structures applied to the independent norming. All four refers to Set, Ordering, Conjunction, and Disjunction; None means that none of these four structures predicts exclusion. These groups are not the predictions of the fitted probabilistic models. Values are equal-context means of item exclusion rates. Each condition contains 852 responses from 213 participants across 16 contexts. The number of distinct trigger–query pairs differs by condition; every context contributes 12, 3, 3, and 12 pairs to the four rows, respectively. The 3,408 responses cover all 480 pairs. These are descriptive means.

linking. Top-p defines a sampled set by cumulative continuation probability mass rather than a fixed number of alternatives. Under the selected boundaries, requiring membership in these sets gives poor exclusion predictions. Disjunction can still predict exclusion through relative ordering when the query is not in the sampled set. Its performance suggests that this second route reduces dependence on the particular membership boundary. The set boundaries are selected using the common training-fold Set objective and are not optimized separately for each hybrid rule.

The FRIDGE context illustrates a constraint that set membership can overcome. Participants excluded juice after "I only have water" and water after "I only have juice" in every observed response for those two pairs. No linking assigned probabilities of .010 and .056, respectively. Set Top-K assigned .762 and 1.000. Ordering assigned .148 and .852: its precedence criterion constrains the two reversed-pair probabilities to sum to one, although human exclusion can be high in both directions. The example shows how membership can support exclusion even when an alternative is unlikely to precede the trigger. It illustrates a constraint of Ordering rather than a general advantage for Set. Set also leaves substantial errors: its prediction for milk is .260, compared with a mean human rate of .872 across the five triggers.

The aggregate gains vary across contexts. Each of the four rules in the Top-K comparison improves log score over No linking in 13 of the 16 contexts, with the remaining contexts differing by model. The training-fold constant-rate reference scores -334.000 on the focus data, above every evaluated model. The structural mappings improve on raw continuation probabilities while leaving substantial room to improve probability fit.

5.2.1 Contextual expectations predict exclusion rankings

No linking yields a mean within-context exclusion Spearman correlation of $\rho = .584$ across the 16 contexts (Figure 3). The four rules in the Top-K comparison have means between .513 and .569. Thus, the improvement in their log scores does not require higher mean rank agreement: neutral contextual expectations already contain information about which alternatives are more frequently excluded, while the explicit mappings improve the probabilities assigned to those judgments.

The ranking information is particularly clear in BAG ($\rho = .874$ for No linking). Continuation probability orders the six tested words as pen, crayons, chalk, makeup, headphones, and cash. Mean human exclusion rates, averaging the five trigger-specific rates for each query, follow the same order. This agreement does not imply that No linking predicts trigger effects: its probability for a query is unchanged when the trigger changes.

The same context illustrates why ranking and probability prediction must be distinguished. After "I only have cash," participants excluded pen, crayons, and chalk in all observed responses (5/5, 2/2, and 5/5, respectively), while No linking assigned probabilities of approximately .0192, .000187, and .0000271.

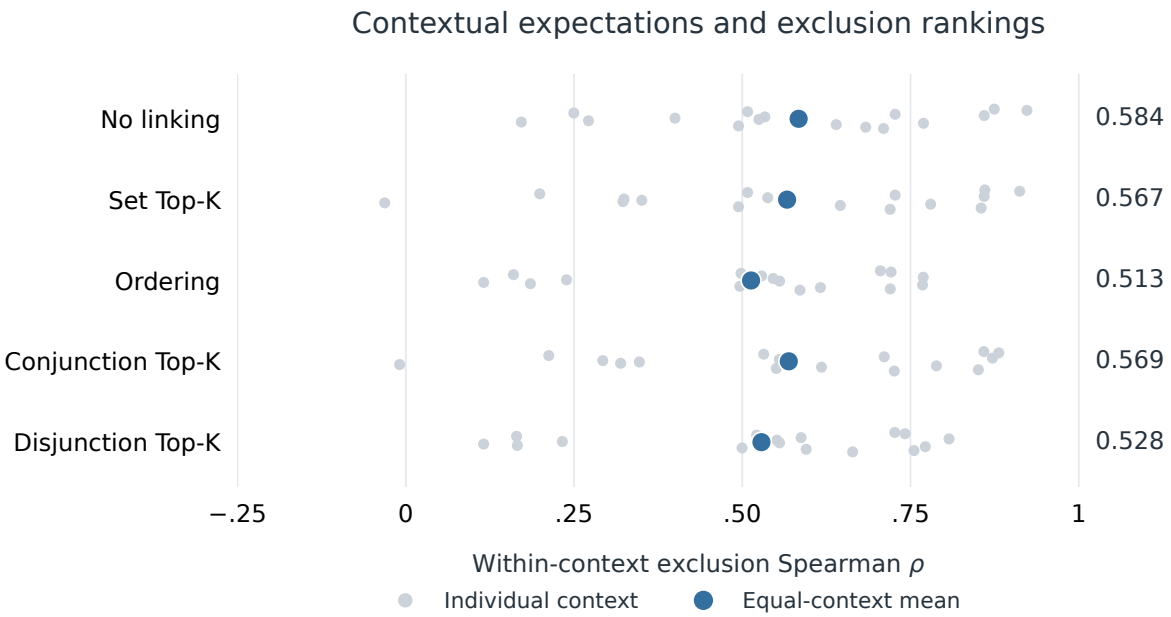


Figure 3: Agreement with human exclusion rankings in the focus experiment. Small gray points show Spearman correlations across all 30 ordered trigger–query pairs within each of the 16 contexts. Large blue points and numerical labels show the equally weighted context means; all 16 correlations are defined for each displayed model. Set, Conjunction, and Disjunction use the selected Top-K boundaries; Ordering is boundary-free. The small vertical offsets separate context points visually. Context variation is descriptive and does not represent confidence intervals.

Those values can preserve an informative relative order while remaining far below the observed exclusion rates. Spearman characterizes the ranking information; log score evaluates the probability predictions.

5.2.2 Linking gains persist when alternative rankings agree

A possible source of linking gains is that the language model misorders the contextual alternatives. We therefore examined whether the log-score improvement persists where the model's ranking of the six tested words agrees closely with their independent human norming order. We retained the original predictions and selected set boundaries without refitting. Figure 4 compares No linking and Set Top-K on the full sample and the two overlapping ranking-agreement subsets.

BAKERY and FRIDGE have perfect word-ranking agreement ($\rho = 1$), providing 60 trigger–query pairs. On these items, No linking has a summed log score of -178.226, compared with -68.169 for Set Top-K and -49.088 for Disjunction Top-K. The mean exclusion-ranking correlations for No linking and Set Top-K are similar, at .891 and .886. Correctly ordering the tested alternatives therefore leaves a further problem of assigning probabilities to their exclusion.

The broader criterion of word-ranking $\rho > .9$ retains seven contexts and 210 pairs. No linking scores -699.143, Set Top-K -374.878, and Disjunction Top-K -225.675. Mean exclusion-ranking correlations for No linking and Set Top-K are again close, at .690 and .688. These descriptive subsets show that the observed log-score gain is not confined to contexts with inaccurate rankings of the tested words. They do not constitute independent replications or establish that the full continuation distribution is correct.

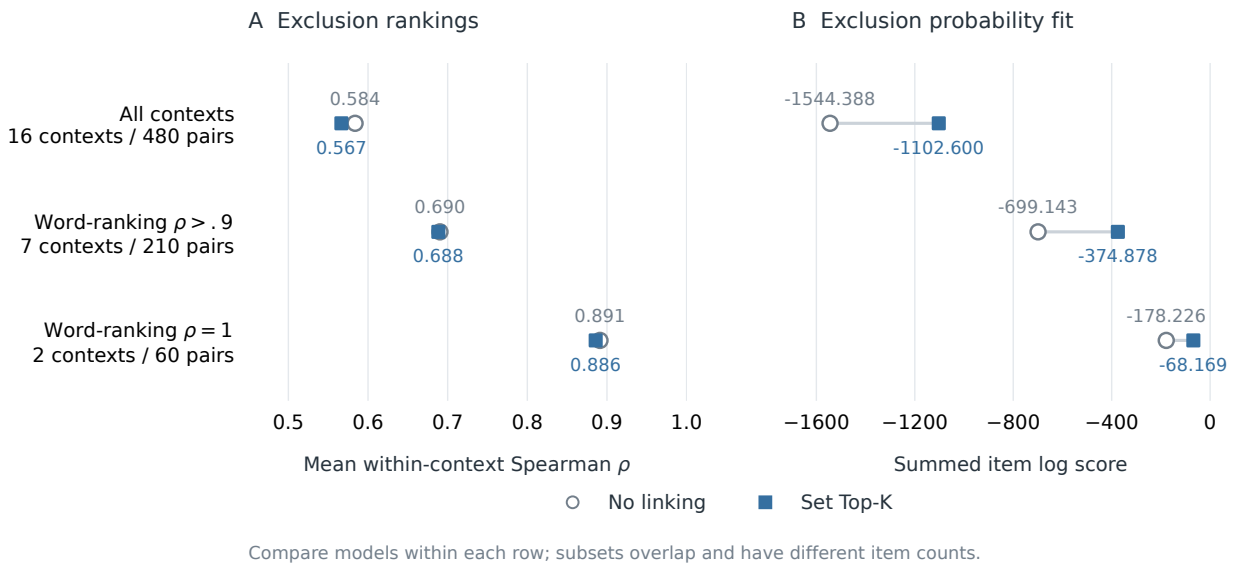


Figure 4: Exclusion prediction when model and human word rankings agree. Panel A shows the equal-context mean Spearman correlation over 30 ordered trigger–query pairs per context. Panel B shows summed item log scores on the same rows. Open circles indicate No linking and blue squares indicate Set Top-K; lines connect the two models within each row. Rows contain the full sample (16 contexts, 480 pairs), contexts with word-ranking $\rho > .9$ (7 contexts, 210 pairs), and contexts with word-ranking $\rho = 1$ (2 contexts, 60 pairs). Word-ranking Spearman compares neutral continuation scores with independent norming ranks for the six tested alternatives. The subsets overlap, and predictions and boundaries are unchanged. Compare log scores between models within a row, because rows have different item counts. Values and connecting lines are descriptive, with no uncertainty intervals or significance tests.

5.3 Contrastive conditioning improves scalar log scores

Across all four Hu et al. (2023) scalar benchmarks, X but not Y improves summed log score over No linking (Table 3). These benchmarks test exclusion of a conventional stronger alternative after a weaker expression. Supplying the weaker expression and an explicit contrast therefore improves the probability predictions in each dataset separately.

The highest-scoring approach varies across datasets. X but not Y leads in Gotzner et al. (2018), Set Top-p in van Tiel et al. (2016), Ordering in Pankratz and van Tiel (2021), and Disjunction Top-K in Ronai and Xiang (2022). Improvements in log score need not coincide with better rank agreement: in Gotzner et al., X but not Y has Spearman $\rho = .043$, compared with .081 for No linking.

The contrast with focus alternatives concerns how the exclusion relation enters the prediction. In each Hu benchmark, supplying the weaker expression and "but not" improves log score over No linking. In the focus experiment, the strongest evaluated probability predictions come from explicit mappings applied to neutral contextual expectations. This comparison constrains how expectations should be used for the evaluated tasks; it does not establish that scalar and focus inferences require different human processing mechanisms.

5.4 Within-condition fit does not capture the contribution of *only*

The Ronai and Xiang (2024) conditions compare the same 60 scales under different QUDs and with or without *only*. At least one explicit rule improves log score over No linking in each condition. The

completed X but not Y predictions also improve log score over No linking in all five conditions, leading the evaluated models in the weak- and strong-QUD conditions. Ranking agreement remains uneven: X but not Y has Spearman $\rho = .497$ in Baseline and .335 in Weak QUD, but only .032 in Strong QUD. Its values are .413 in *Only* and .060 in Strong QUD + *only*.

The matched *only* comparisons reveal a limitation that within-condition fit does not resolve. Adding *only* increases mean human exclusion from .387 to .684 without an explicit QUD, and from .613 to .886 with a strong QUD. Every evaluated model, including X but not Y, nevertheless produces identical predictions for matched items with and without *only*. The implementation omits *only* from the scoring prompts and applies the same mapping in both conditions. Consequently, differences in log score across these matched conditions reflect the changed human responses rather than a modeled response to *only*.